

Manual for Agentic Trading Lab

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Overview

Agentic Trading Lab is an interactive playground for research and educational purposes. for exploring LLM-based trading agents. It accompanies our systematic survey of around 200 papers on agentic trading, which shows that the field is moving beyond simple price prediction toward agents that can retrieve information, reason, make decisions, and trade.

The Lab is motivated by the long-standing appeal of paper-trading competitions in algorithmic trading. In the algorithmic trading era, these competitions allowed students, researchers, and practitioners to experiment with trading strategies in a low-risk but realistic setting. Agentic Trading Lab aims to bring this spirit into the agentic trading era: users can create an LLM-based trading agent, deploy it into a contest-style environment, and observe how it behaves through backtesting and paper trading.

Through the Lab, students, researchers, and practitioners can customize trading agents, run backtests and paper-trading experiments, inspect decision logs, and compare performance metrics. The goal is to help users understand not only whether an agent can generate returns, but whether it is AI-ready for trading deployment under realistic constraints.

Purpose and Community Impact

The purpose of **Agentic Trading Lab** is to make agentic trading research more accessible by providing a browser-based environment for exploring the full trading-agent workflow, from backtesting to paper trading and performance evaluation.

The Lab is designed to help users move beyond the Alpha Illusion. By connecting backtesting, paper trading, decision logging, risk monitoring, and leaderboard evaluation, the Lab helps users understand how LLM-based trading agents behave in and beyond historical simulation.

Specifically, the Lab aims to:

1. **Lower the barrier to entry**
Provide students and new researchers with a ready-to-use interface for backtesting, paper trading visualization, and leaderboard comparison.
2. **Bridge the alpha-deployment gap**
Help users understand how execution, transaction costs, market timing, drawdown, and operational constraints affect real-world trading performance.
3. **Evaluate AI-readiness for trading deployment**
Test whether LLM-based agents can move from historical simulation toward deployment-aware trading under realistic market and operational constraints.
4. **Expose the defense side of trading agents**
Introduce users to deployment-critical components such as portfolio risk, P&L logging, decision records, drawdown monitoring, and governance-oriented evaluation.

5. **Support academic trading competitions**

Provide a common interface where participants can submit trading decisions and compare risk-adjusted realized performance over a fixed competition window.

Key Features

- **Trading Agents Powered by LLMs**

Agentic Trading Lab supports LLM-based trading agents that generate BUY, SELL, and HOLD decisions with natural-language reasoning.

- **Prompt-Based Trading**

Users can describe trading ideas in plain text and turn them into structured trading workflows for testing.

- **Interactive Backtesting Interface**

Users can simulate trading agents on historical market data with selected time ranges, assets, and agent configurations.

- **Decision Log Visualization**

The Lab records each agent decision with its action, price, and reasoning so users can connect rationale with performance.

- **Paper Trading View**

Users can observe a trading agent's live-market trading results, including portfolio value, positions, trades, and account summary.

- **Competition Leaderboard**

The Leaderboard compares team submissions by realized trading performance over a fixed competition window.

QuickStart

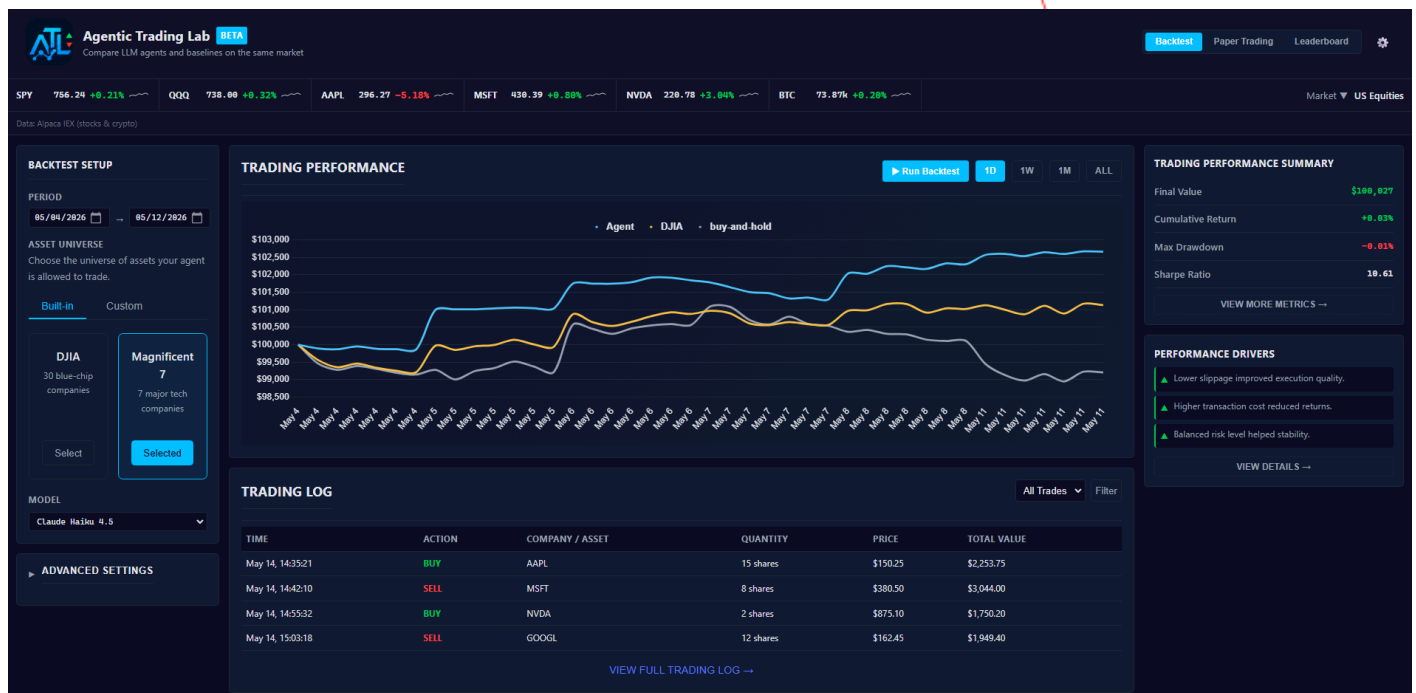
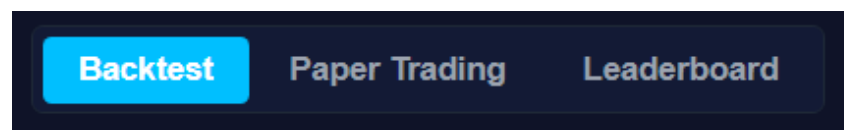
1). Run Agentic Trading Lab on your browser:

<https://agentic-trading-lab.vercel.app/>

No installation is required.

2). Choose one of the main modes from the top navigation bar:

- **Backtest:** run historical trading simulations and compare the agent with market baselines.
- **Paper Trading:** view simulated real-market trading results from an **Alpaca paper trading account**, including portfolio value, open positions, and recent trades.
- **Leaderboard:** view and compare team submissions for the **SecureFinAI Contest 2026 Task V**.



User Scenario 1: Try Backtesting

This scenario helps users run their first historical backtest and understand the basic workflow of Agentic Trading Lab. A backtest allows users to see how an LLM-based trading agent would have performed during a selected historical period.

Steps

1. Open the Agentic Trading Lab website:
<https://agentic-trading-lab.vercel.app/>
2. Select **Backtest** from the top navigation bar.
3. In the **Backtest Setup** panel, choose a time period by selecting a start date and an end date.
4. Under Task Type, select Portfolio Management.
5. Select the assets you want the agent to trade. For a first test, users can begin with a small group of familiar stocks, such as AAPL and MSFT.
6. Choose the model used by the trading agent.
7. Click **Run Backtest**.
8. Wait for the simulation to complete.
9. Review the output:
 - **Trading Performance**: portfolio value over time
 - **Baselines**: DJIA and buy-and-hold comparisons
 - **Decision Log**: agent actions, prices, and reasoning

Learning Outcome

By completing this scenario, users will understand the basic workflow of Agentic Trading Lab: selecting a backtest period, choosing assets, running the simulation, and reviewing the agent's performance and

BACKTEST SETUP

PERIOD

05/04/2026 → 05/12/2026

ASSET UNIVERSE

Choose the universe of assets your agent is allowed to trade.

Built-in

Custom

DJIA

30 blue-chip companies

Select

Magnificent 7

7 major tech companies

Selected

MODEL

Claude Haiku 4.5

TRADING PERFORMANCE

Run Backtest 1D 1W 1M ALL

Agent DJIA buy-and-hold

Date	Agent	DJIA	buy-and-hold
May 4	\$99,500	\$99,500	\$99,500
May 5	\$101,000	\$100,000	\$99,000
May 6	\$101,500	\$100,500	\$99,500
May 7	\$101,000	\$100,000	\$99,000
May 8	\$102,000	\$100,500	\$99,500
May 9	\$102,500	\$101,000	\$99,000
May 10	\$102,500	\$101,000	\$99,000
May 11	\$102,500	\$101,000	\$99,000

User Scenario 2: Evaluate Trading Performance

This scenario helps users understand how to interpret the results of a backtest. Instead of only looking at the performance curve, users learn how to evaluate the agent using standard trading metrics.

Steps

1. Run a backtest using the steps from **User Scenario 1**.
2. After the backtest completes, locate the **Trading Performance Summary** panel on the right side of the page.
3. Review the main metrics:
 - **Final Value**: the portfolio value at the end of the backtest.
 - **Cumulative Return**: the total percentage gain or loss during the selected period.
 - **Maximum Drawdown**: the largest drop from a previous portfolio high, showing downside risk.
 - **Sharpe Ratio**: a risk-adjusted return measure; higher values usually indicate better return relative to volatility.
4. Compare the metrics with the equity curve and decision log to understand both performance and trading behavior.
5. Use the **Decision Log** to connect the metrics with the agent's behavior. For example, frequent trading may increase transaction costs or lead to larger losses during volatile periods.

Learning Outcome

By completing this scenario, users will learn how to move from visual observation to performance evaluation. They will understand how final value, cumulative return, maximum drawdown, and Sharpe ratio help evaluate



🇺🇸 TRADING PERFORMANCE SUMMARY	
Final Value	\$100,027
Cumulative Return	+0.03%
Max Drawdown	-0.01%
Sharpe Ratio	8.59
VIEW MORE METRICS →	

